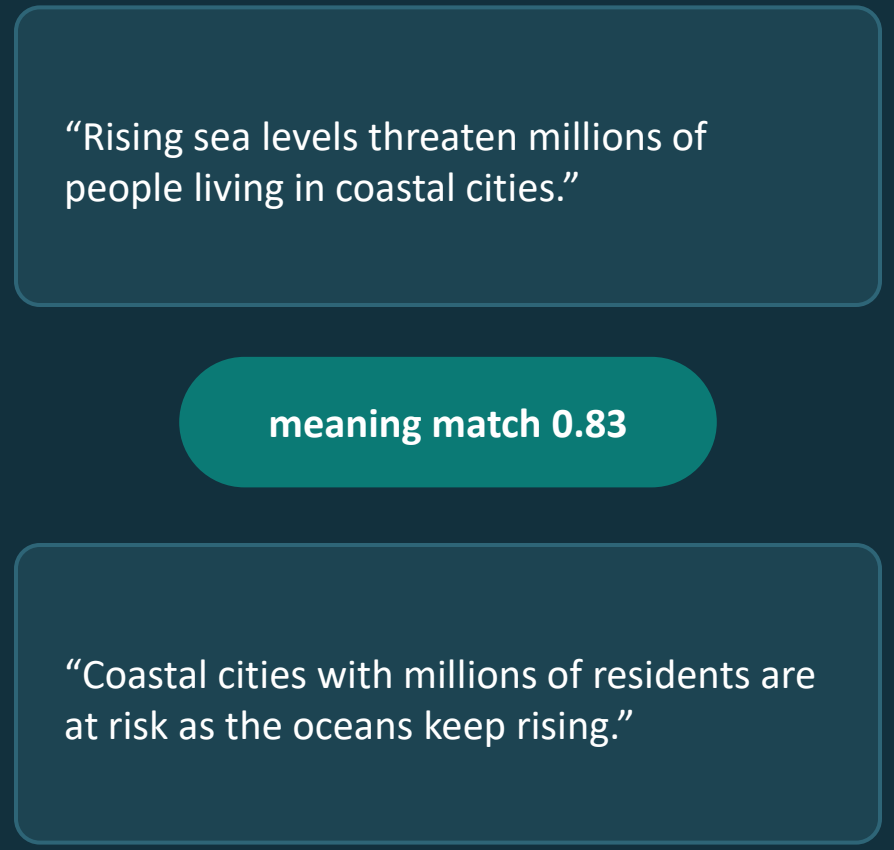
EXPERIENTIAL LEARNING | GROUP 4

Semantic Similarity and Plagiarism Detection

Finding passages that mean the same thing, and showing why that is not the same as copying.

Oishani Majumder 2330316
Shubhradeep Paul 2330337



A copied idea does not need copied words

"Rising sea levels threaten millions of people living in coastal cities."

same idea

"Coastal cities with millions of residents are at risk as the oceans keep rising."

Humans see it instantly. A checker that only looks for matching words does not.

0.16

word overlap
(looks different)

0.83

meaning similarity
(looks the same)

And the catch: two students can independently write the same fact. Similar meaning is a reason to look closer, not proof of copying.

What we set out to do

1

Find similar passages

Turn every sentence into numbers that capture its meaning, then compare two documents sentence by sentence.

2

Tell copying from rewording

Add a word-overlap check so the report says “likely copied” or “paraphrase”, not just “similar”.

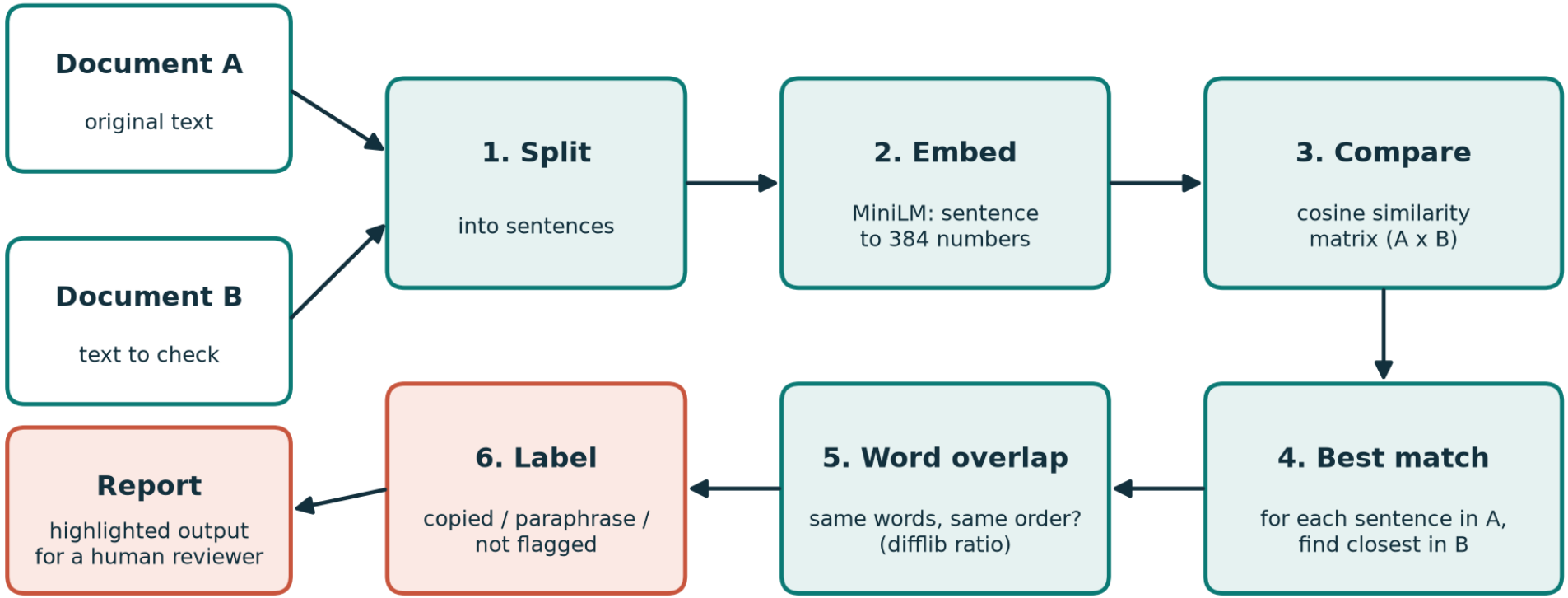
3

Be honest about the result

Test the method on public benchmarks and show clearly why a high score is not proof of plagiarism.

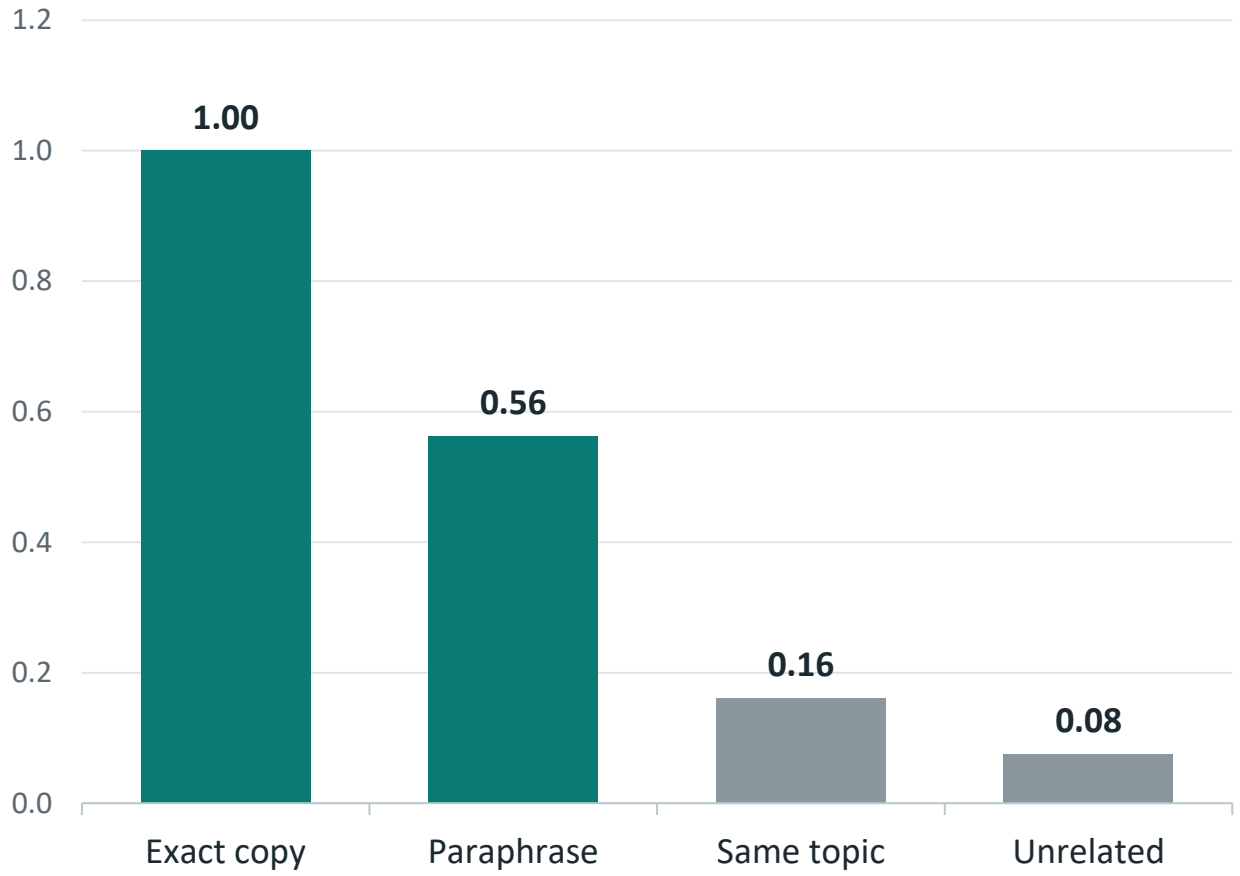
Task 4 from the assignment list. Pretrained model, no training needed, runs on a free Colab CPU.

How it works in six steps



Nothing is trained. The model is pretrained; our own work is the comparison logic around it.

Sentences become numbers, meaning becomes closeness



Compared with: "The cat sat on the mat."

384 numbers describe each sentence

Angle between two lists = cosine similarity

1.0 same meaning

~0 unrelated

Model: all-MiniLM-L6-v2 (Sentence Transformers)

Models, data and tools

Model

all-MiniLM-L6-v2

About 22M parameters, ~90 MB

Output: 384 numbers per sentence

huggingface.co/sentence-transformers/all-MiniLM-L6-v2

Datasets

STS-B: 1,379 pairs with human similarity scores

MRPC: 408 pairs labelled paraphrase or not

Used for testing only, no training

Hugging Face: stsb, glue (mrpc)

Tools

Python in Google Colab, CPU is enough

sentence-transformers, datasets

scikit-learn: TF-IDF baseline, accuracy, F1

SciPy, pandas, matplotlib, difflib

Two signals: meaning and wording

Every sentence in A is matched to its closest sentence in B, then labelled:

Likely copied	**Meaning ≥ 0.65 and words ≥ 0.75**	Same idea in nearly the same words. Could still be a quotation.
Paraphrase / same idea	**Meaning ≥ 0.65 and words < 0.75**	Same idea, new words. Could be copying or independent writing.
Not flagged	**Meaning < 0.65**	No close match in the other document.

Thresholds are settings, not laws: tuned on our samples and easy to change.

Demo: what the report looks like

Likely copied

A1: Climate change refers to long-term shifts…

meaning 1.00 | words 1.00

Paraphrase / same idea

A2: Burning fossil fuels such as coal, oil and gas…

meaning 0.70 | words 0.18

Paraphrase / same idea

A3: Renewable energy sources like solar and wind…

meaning 0.73 | words 0.08

Not flagged

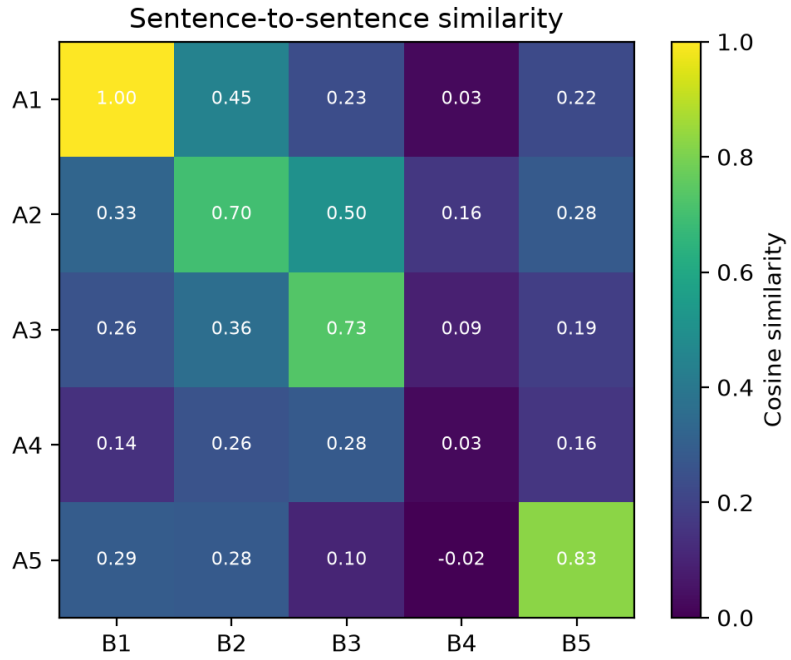
A4: Many countries have pledged to reach net-zero…

meaning 0.28 | words 0.23

Paraphrase / same idea

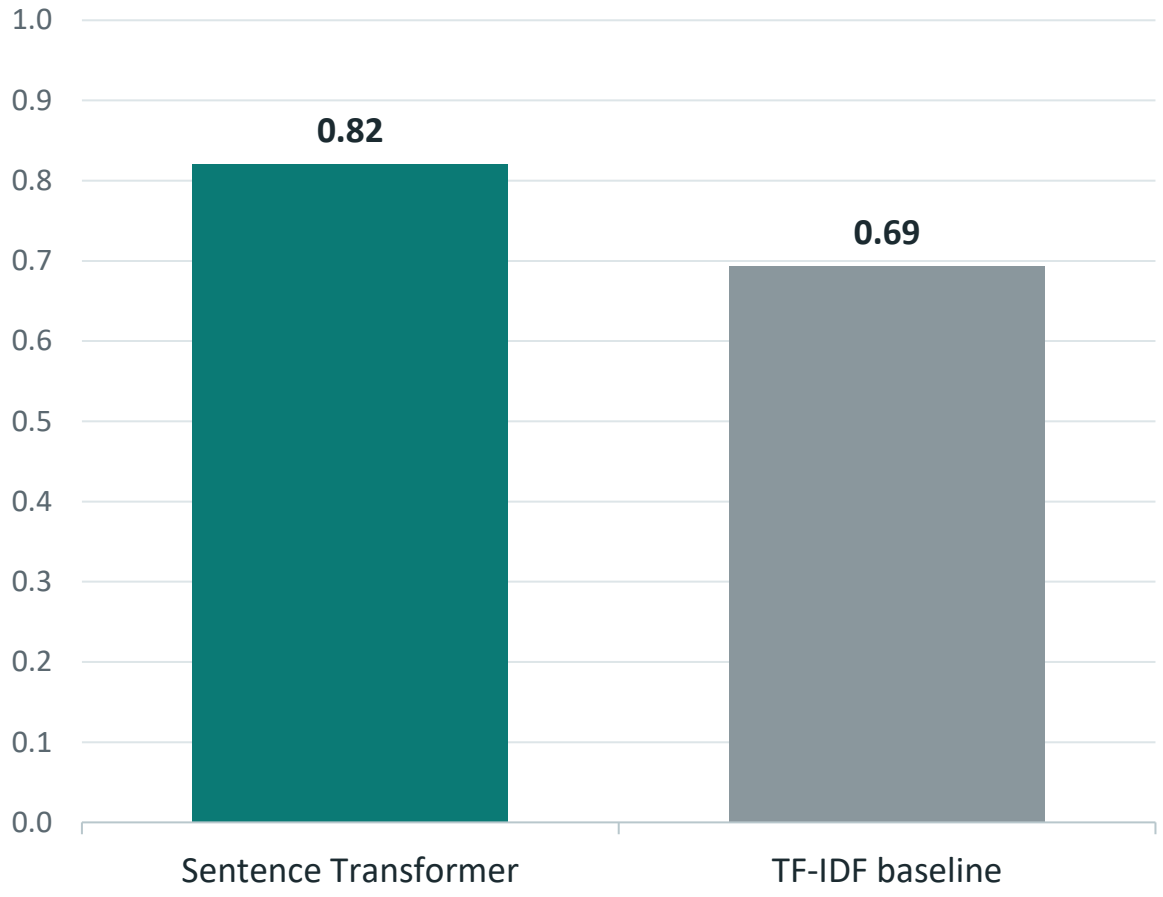
A5: Rising sea levels threaten millions of people…

meaning 0.83 | words 0.16



Rows: sentences of A. Columns: sentences of B. Bright = close in meaning.

Does the model agree with humans?



STS-B test set, 1,379 sentence pairs

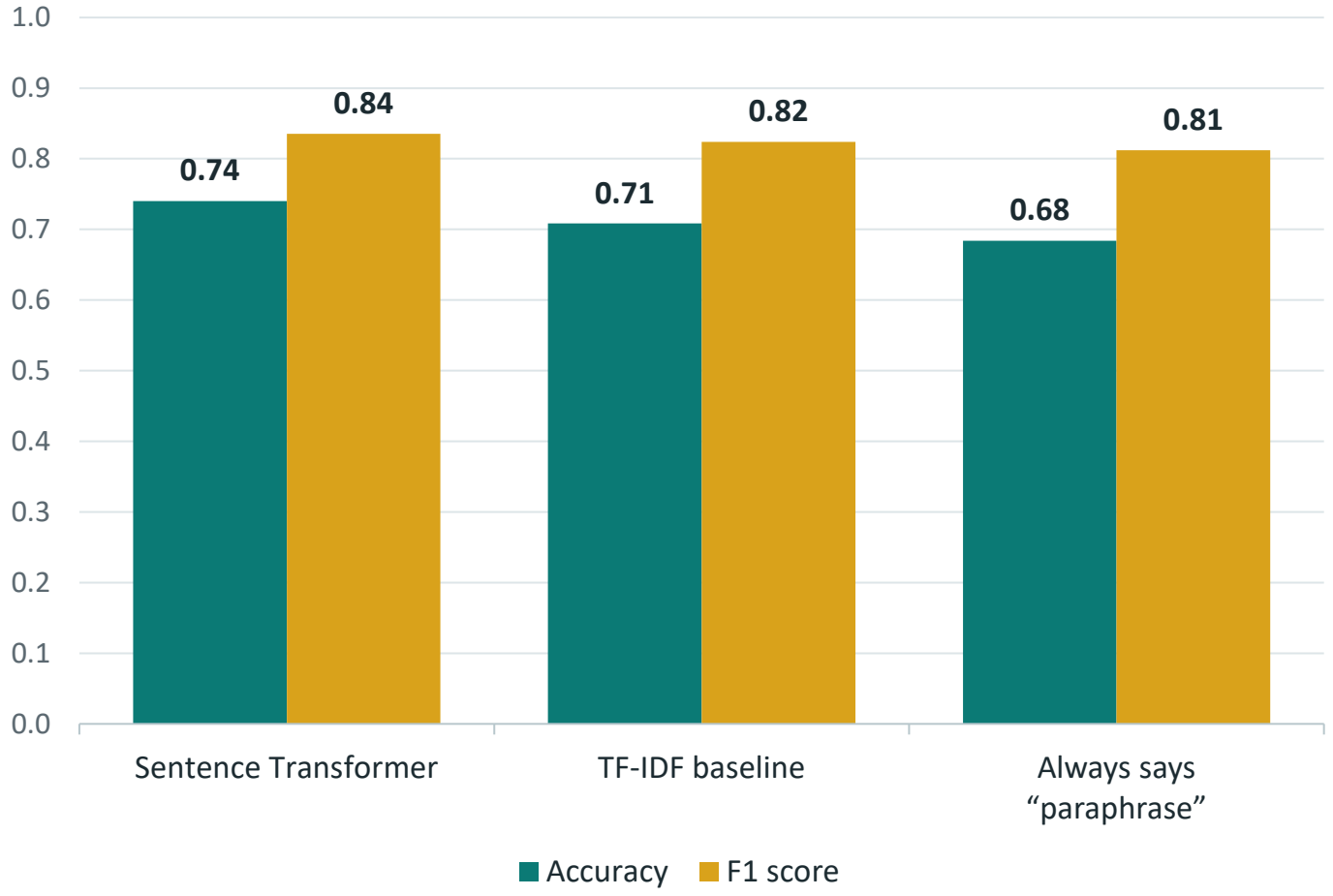
0.82

Spearman correlation with human scores

TF-IDF, which only counts shared words, reaches 0.69.

In plain words: when people say one pair is more similar than another, the model usually agrees, and more often than word matching does.

Spotting paraphrases: better, but only modestly



Accuracy 74.0% vs 70.8% for TF-IDF. Better, but the gap is small.

68% of pairs are paraphrases, so even "always yes" scores well.

Threshold tuned on the same pairs, so figures are a little optimistic.

Lesson: a threshold is a trade-off, so we show scores, not just a verdict.

Accuracy at a glance

0.82

STS-B agreement with humans

TF-IDF: 0.69

74.0%

MRPC accuracy

TF-IDF: 70.8%

94.4%

Labeller accuracy (17 of 18)

Chance level: 33.3%

Labeller results (rows: true label, columns: predicted)

	Likely copied	Paraphrase / same idea	Not flagged
Likely copied	**6**	0	0
Paraphrase / same idea	0	**5**	1
Not flagged	0	0	**6**

The one miss: a real paraphrase scored under the 0.65 threshold, so it was not flagged.

Similar is not the same as plagiarised

0.92

cosine similarity, and nobody copied

“Water boils at 100 degrees Celsius at sea level.”

“At sea level, water reaches its boiling point at 100 degrees Celsius.”

A human must still check

1. **Was the source cited?**
   A properly credited passage is not plagiarism.
2. **Is it a quotation?**
   Quoted text will always match.
3. **Is it common knowledge?**
   Facts and definitions get stated alike.
4. **Was it written independently?**
   Two people can reach the same sentence.

Limits, and where we would take it next

Limitations

Thresholds depend on the kind of text

Short paraphrases can slip under the threshold (cat and mat: 0.56)

Only the closest sentence is compared

English only; two documents at a time, no web search

Our own test set is small

Next steps

Multilingual model for Hindi and Odia

Compare windows of several sentences

Re-rank borderline cases with a cross-encoder

Test on a real plagiarism corpus (PAN)

Upload page for PDF and Word files

What we learned

0.82

agreement with humans on STS-B, vs 0.69 for word matching

2 signals

meaning plus wording separate “copied” from “paraphrased”

Human first

a high score is a pointer for review, never a verdict

Thank you. Questions?

TEAM

Oishani Majumder 2330316

Models, datasets, evaluation, results

Shubhradeep Paul 2330337

Comparison pipeline, GitHub, slides

CODE, REPORT AND VIDEO

github.com/<your-username>/<your-repo-name>